Jérémiah Emenike CHUKWUEKE

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Research interests: Scientific machine learning · Process/physics-informed neural networks · Spatio-temporal and geospatial data · Trustworthy ML · NLP for low-resource languages

EDUCATION

M.Sc. in Artificial Intelligence and Big Data, Collège de Paris Supérieur, Lomé, Togo · 2022–2024

- Thesis: *Automating Candidate Preselection in Online Recruitment*. Advisor: Dr. Amana Maniyassouwe

B.Sc. in Mathematics and Computer Science, Catholic University of West Africa (UCAO), Academic Unit of Togo · 2018–2022

High School Diploma (Mathematics & Physics), Collège Protestant, Lomé · 2018

RESEARCH EXPERIENCE

Master's thesis research, Collège de Paris Supérieur · 2023–2024

- Designed and implemented an end-to-end NLP pipeline that ranks résumés against job descriptions. It covers PDF text extraction (PyMuPDF), preprocessing, Word2Vec embeddings and cosine-similarity ranking.
- Built an evaluation protocol against recruiter ground truth (precision and recall on the top k, order agreement, NDCG). Results on three job postings: **100% precision and an NDCG of 0.99 on the top 3 candidates**.
- Reduced estimated screening time for 100 résumés from about 5 hours to about 20 seconds (about 99% faster).
- Analyzed the system's limitations: algorithmic bias and fairness in hiring, static versus contextual embeddings, and data-privacy ethics.
- Deployed the system as a Flask web application for non-technical users.

Road sign detection with convolutional neural networks, Data Science internship · 2022–2024

- Improved road sign detection using CNNs.

PROFESSIONAL EXPERIENCE

Data Engineer, Togo AI Lab, Ministry of Public Service Efficiency and Digital Transformation, Lomé · March 2025–present

- Designed and developed the national Big Data architecture (Apache NiFi, Hadoop, Hive) used for the National Digital Observatory case study.
- Developed **geospatial data pipelines** (Python, PostGIS, QGIS, QField).
- Contributed to the architecture of the **Togo Rural Roads Database**.
- Monitored data flows and government digital platforms with Grafana.

Data Science Intern, Collège de Paris Supérieur & Chukwu Store · September 2022–November 2024

- Worked on AI-based automation of candidate pre-screening (Master's thesis) and on CNN-based road sign detection.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Bash
- **Machine learning / deep learning:** PyTorch, TensorFlow, scikit-learn, Gensim (Word2Vec), CNNs, NLP
- **Data engineering:** ETL/ELT, Apache Airflow, Kafka, NiFi
- **Big data:** Spark, Hadoop, Hive, data partitioning
- **Geospatial:** PostGIS, QGIS, QField
- **Tools:** Flask, Streamlit, Grafana, Git
- **Languages:** French, English

REFERENCES

- James Adeola, Collège de Paris Supérieur, Jadeola@collegedeparis.fr
- Essowaba Ahouloumi, Collège de Paris Supérieur, eahouloumi@collegedeparissuperieur.fr
- Frank Assou, Collège de Paris Supérieur, F.assou@keyce-informatique.fr